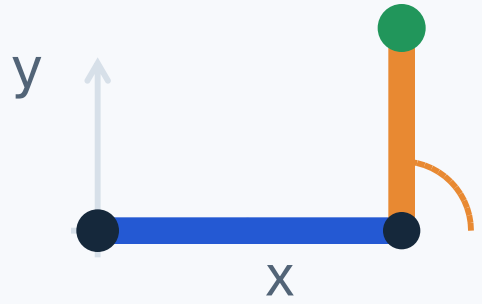
Gazebo, ROS & kinematics

Build a simulated world. Predict motion. Verify the result.

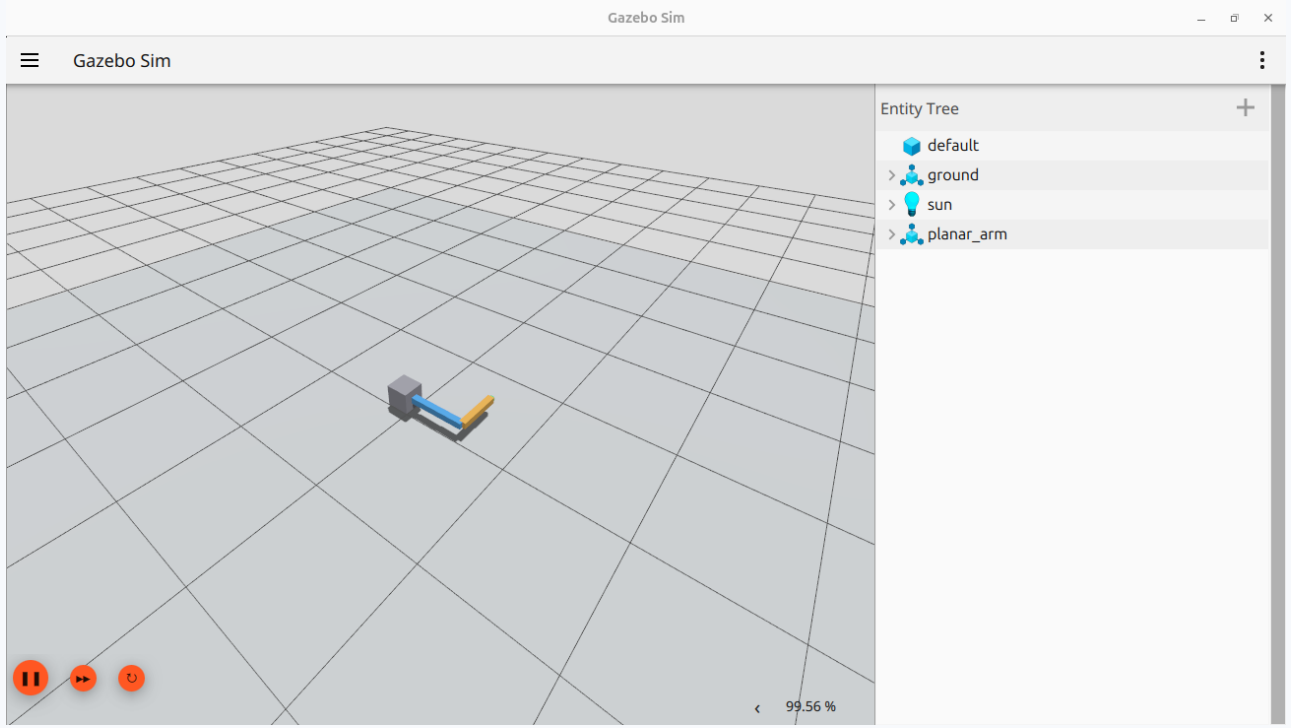
Understand Gazebo | **Connect ROS** | **Practice kinematics**



20 min explanation
25 min code, demo & questions

What is Gazebo?

A robotics simulator: a modeled system evolving in simulation time.



World + environment

Robot models

Physics + controllers

Simulation time

Optional sensors

Why simulate robots?

Test ideas early and observe them under repeatable conditions.

Before hardware

Develop early.

Repeat a setup

Compare experiments.

Try commands

Observe virtual motion.

Inspect results

Read joints, frames, and time.

Results depend on the model. Validate later on hardware.

Why Gazebo fits this course

ROS integration and reusable configuration keep attention on robotics.

ROS integration

Launch, spawn, bridge messages.

Configurable models

Robot, world, and topic mapping.

Reusable systems

Physics, control, sensors.

GUI or headless

Same model and physics.

Gazebo, ROS, and RViz

Each has a different role in the same robotics workflow.

Gazebo

Simulates motion
Produces measurements

ROS

Communicates data
Runs robot logic

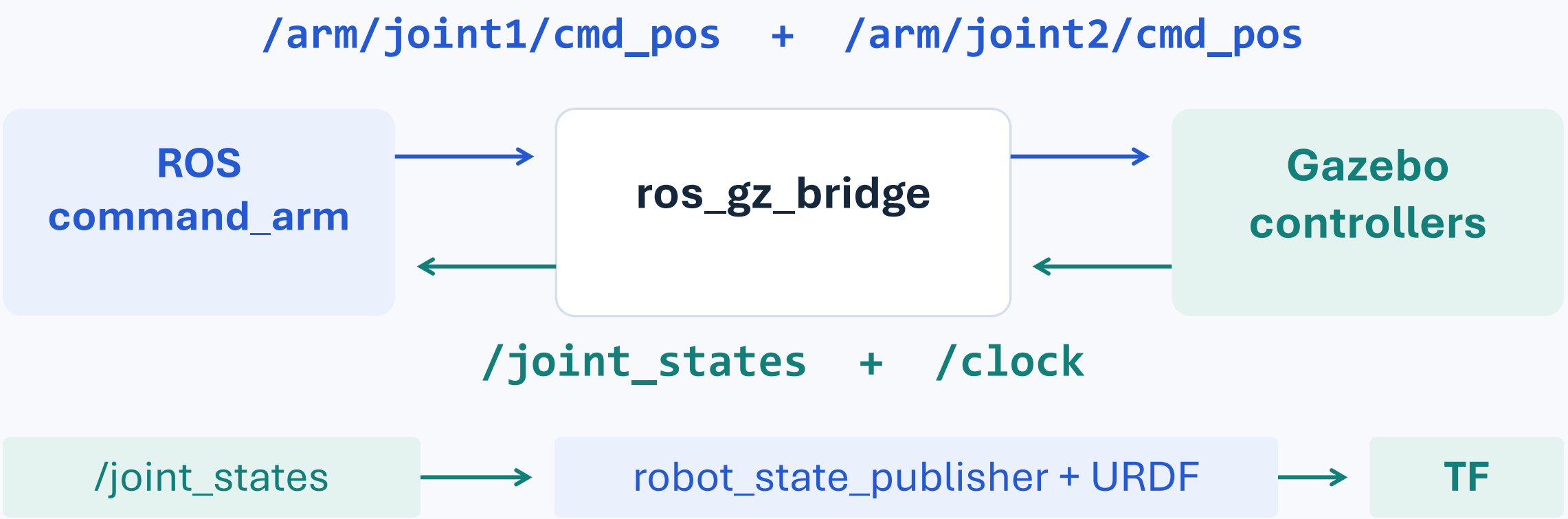
RViz

Visualizes ROS data
Optional in this demo

Simulation ↔ ROS data → visualization

The ROS–Gazebo bridge

Explicit message mappings connect two communication systems.



The workflow

Describe → launch → connect → command → inspect → compare.



Configure robot and world

Small descriptions assemble a useful simulation.

URDF · joint geometry

```
<origin xyz="0.6 0 0"/>
<axis xyz="0 0 1"/>
```

SDF · world physics

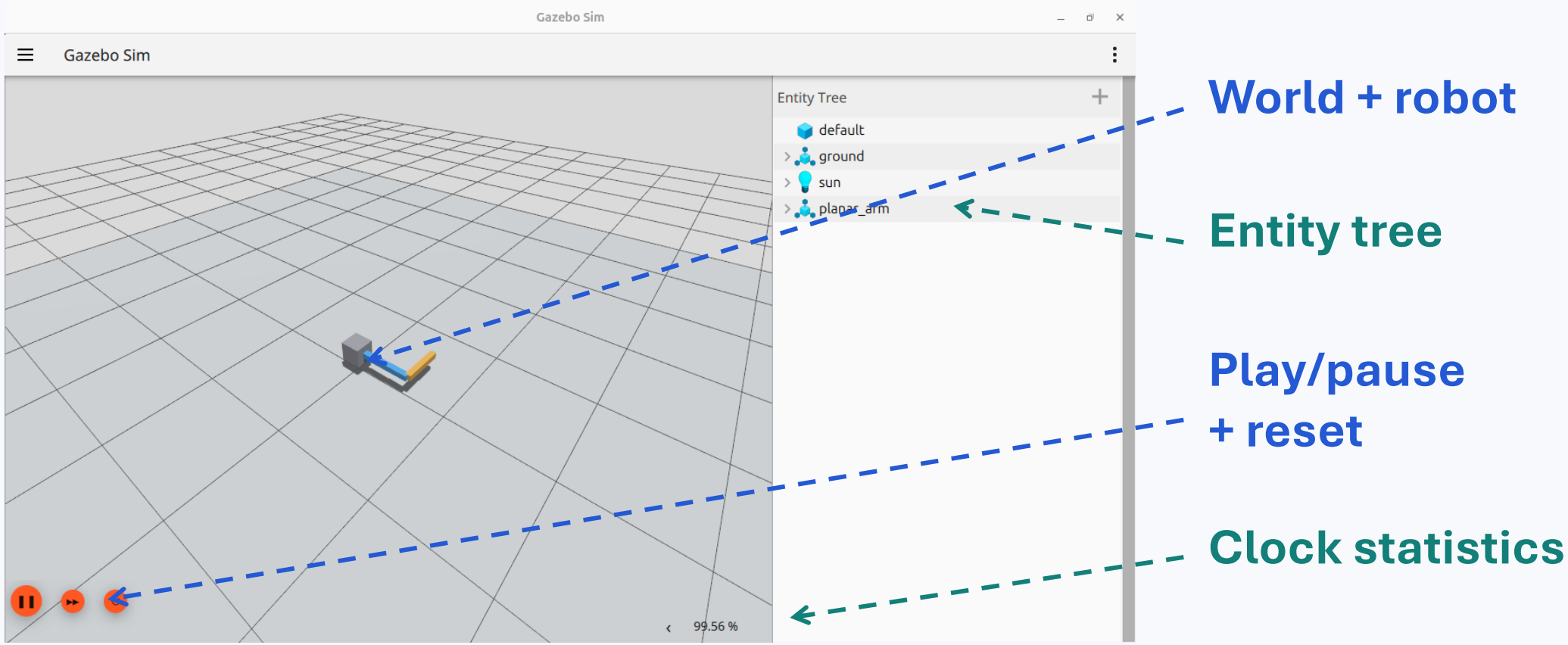
```
<gravity>
  0 0 -9.81
</gravity>
```

bridge.yaml

```
direction: ROS_TO_GZ
```

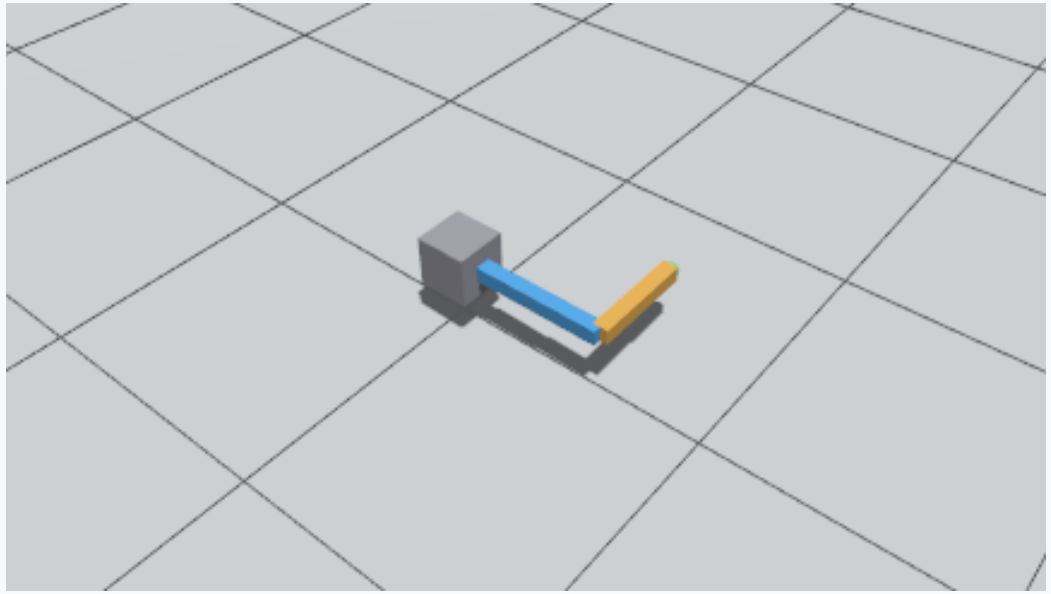
Read the simulation

The view, robot state, and simulation clock are separate.



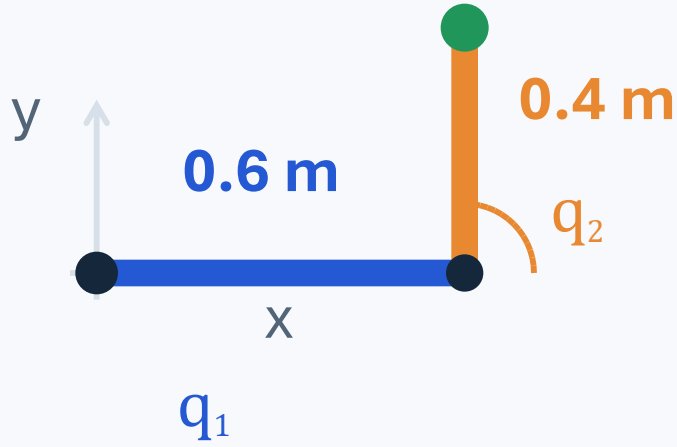
Our two-joint arm

Zero points along +x; both joints rotate about +z.



The existing Gazebo demo

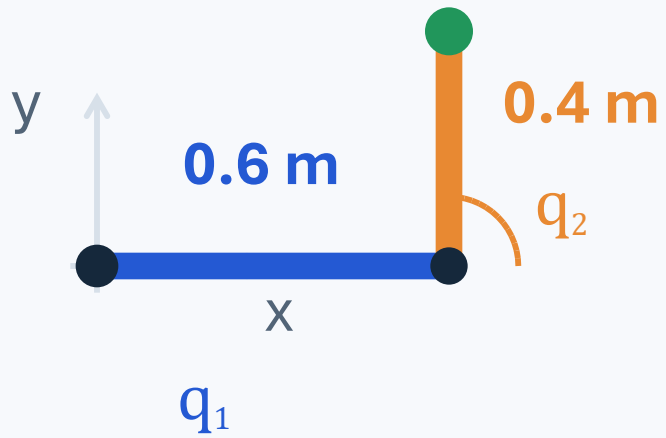
Top view · base_link



q_2 is relative to link1.

Forward kinematics

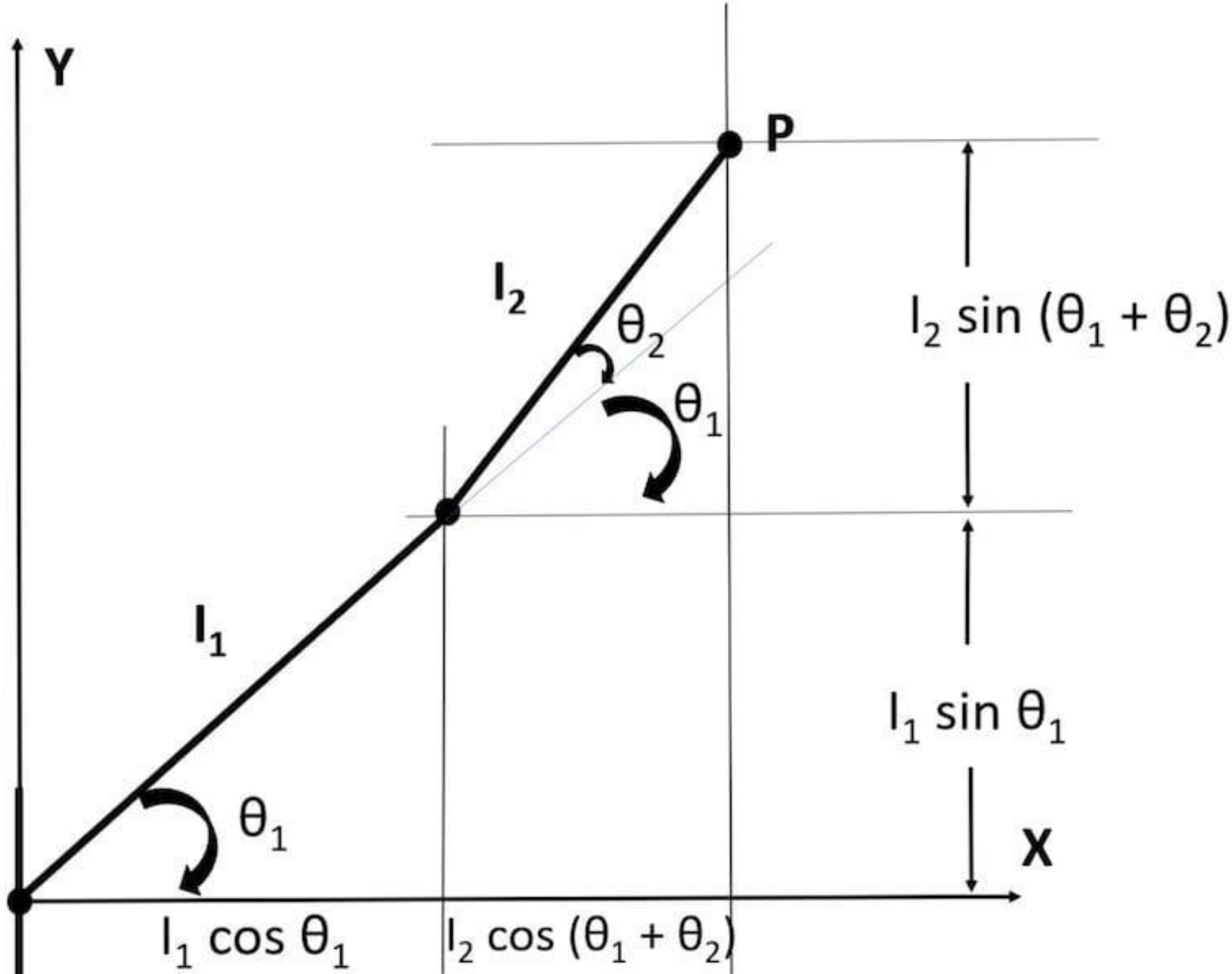
Joint angles → tool position in base_link.



$$x = L_1 \cos(q_1) + L_2 \cos(q_1+q_2)$$

$$y = L_1 \sin(q_1) + L_2 \sin(q_1+q_2)$$

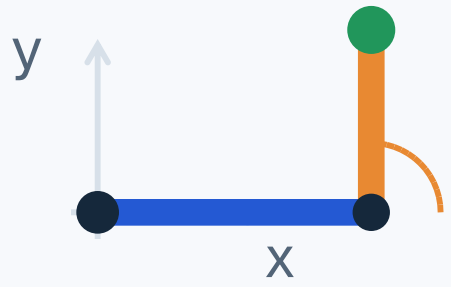
$(q_1, q_2) = (0, \pi/2)$ → $(x, y) = (0.6, 0.4)$ m


Y
P
l_2
θ_2
θ_1
$l_2 \sin(\theta_1 + \theta_2)$
l_1
$l_1 \sin \theta_1$
θ_1
X
$l_1 \cos \theta_1$
$l_2 \cos(\theta_1 + \theta_2)$

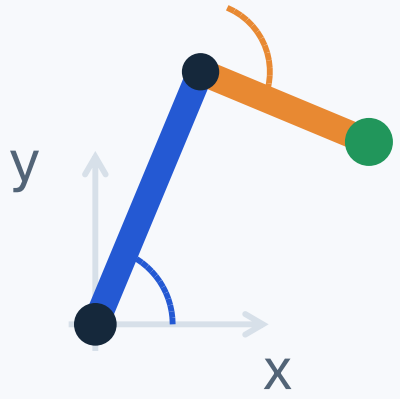
Inverse kinematics

One tool target can have two arm configurations.

Positive q_2



Negative q_2



Same target: (0.6, 0.4) m

$$c_2 = (x^2+y^2-L_1{}^2-L_2{}^2)/(2L_1L_2)$$

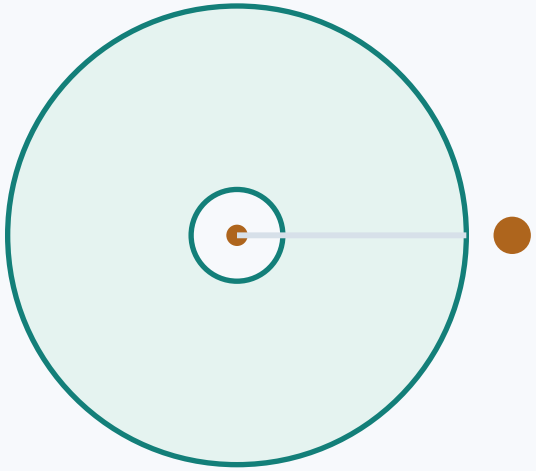
$$q_2 = \pm\operatorname{acos}(c_2)$$

$$q_1 = \operatorname{atan2}(y,x) - \operatorname{atan2}(L_2 \sin q_2,\ L_1+L_2 \cos q_2)$$

Verify what happened

A requested target is a prediction; measured feedback checks it.

Reject (0,0) and (1.2,0).



0.2 ≤ radius ≤ 1.0 m

Measured joints

↓

FK + end-effector TF

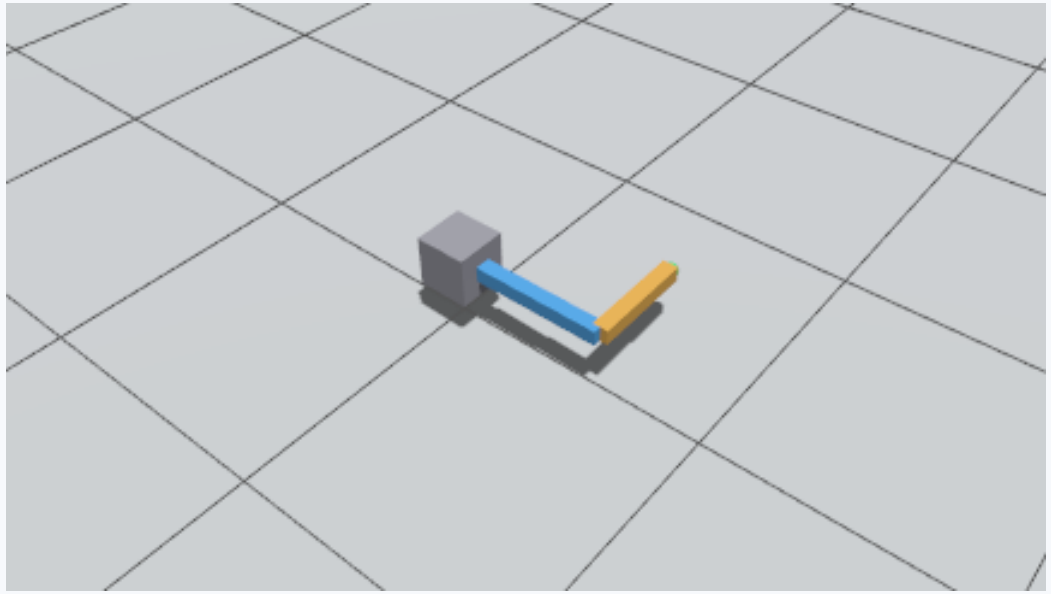
↓

Compare with target

IK ≠ collision-free motion planning

Live demo: predict, command, inspect

Keep the simulated robot and its measured state connected.



Predict → command → measure

8 min · interface + joints

10 min · target + two branches

7 min · rejection + checkpoint

OPTIONAL / DEEPER DISCUSSION

Forward kinematics as transforms

Column vectors, right-handed rotations, and joint-relative frames.

$$T = \mathrm{RotZ}(q_1)\ \mathrm{TransX}(L_1)\ \mathrm{RotZ}(q_2)\ \mathrm{TransX}(L_2)$$

$$\begin{bmatrix} \cos\alpha & -\sin\alpha & 0 & x \\ \sin\alpha & \cos\alpha & 0 & y \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

R: rotation (3×3)
t: origin offset (3×1)

$$T = [R\ \ t;\ 0\ \ 1]$$

$$\alpha = q_1 + q_2$$

OPTIONAL / DEEPER DISCUSSION

Derive the inverse kinematics

Cosine rule determines the elbow; atan2 determines the shoulder.

$$x^2+y^2 = L_1^2+L_2^2+2L_1L_2\cos(q_2)$$

$$q_2 = \pm\text{acos}((x^2+y^2-L_1^2-L_2^2)/(2L_1L_2))$$

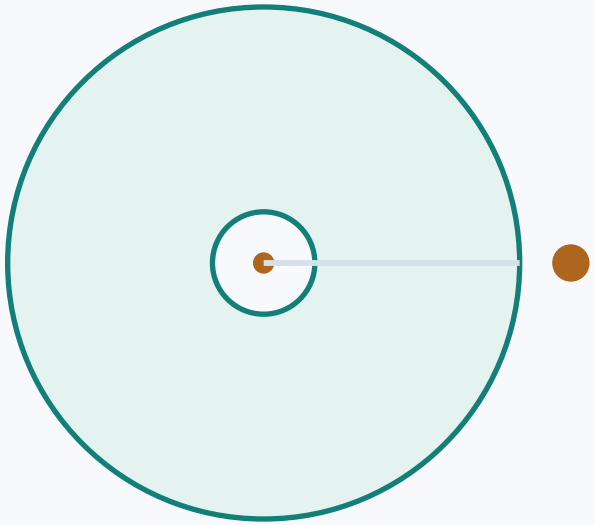
$$q_1 = \text{atan2}(y,x) - \text{atan2}(L_2\sin q_2,\ L_1+L_2\cos q_2)$$

(0.6,0.4): (0,1.570796) or (1.176005,−1.570796) rad

OPTIONAL / DEEPER DISCUSSION

Workspace and singular boundaries

Geometric reachability does not guarantee feasible motion.



0.2 ≤ radius ≤ 1.0 m

$$\det J = L_1 L_2 \sin(q_2)$$

Singular at $q_2 = 0$ or $\pm\pi$.

Joint limits and obstacles add constraints.

OPTIONAL / DEEPER DISCUSSION

Bridge and configuration checks

Trace commands, feedback, and clock separately.

No motion

Pause → bridge → controller

No feedback

Topic mapping + joint names

No TF

Measured joints + URDF

Clock stopped

Check /clock and play/pause

Full restart restores the baseline. GUI reset is different.